MODULE 33 · DOMAIN 6 - EVALUATION AND MONITORING

Metrics that pick models and fit scenarios

Choosing a model's size and architecture from the numbers, and choosing which numbers to watch for each kind of application.

SECTION 6 of 6 EXAM WEIGHT 12% MODULE 33 of 40

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Use the right thermometer for the patient. A metric is only useful if it measures the failure this application would actually suffer.

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Five kinds of quantitative metric

FAMILY	EXAMPLES
Quality	Task metric such as accuracy, judge pass rates
Speed	Time to first token, p50 and p95 latency, tokens per second
Cost	Cost per request, or per thousand requests
Reliability	Error rate, rate-limit rejections, timeouts
Safety	Safety judge pass rate, guardrail trigger rate

ON THE EXAM

Model choices trade these against each other. A choice that wins on one family often loses on another.

Average latency, or the 95th percentile?

AVERAGE, OR P50

The typical request

- Hides a slow minority
- Looks fine while users complain
- Useful for capacity planning
- Not what users remember

P95, OR P99

The slow tail

- 95% of requests are faster than this
- Captures the waits users notice
- What latency targets should use
- Reveals spikes and queueing

THE TIEBREAKER

When users complain but the average looks fine, look at the p95. Users feel the tail, not the mean.

A dense model, or mixture-of-experts?

DENSE

Every parameter, every token

- All weights used for each token
- Cost and latency grow with size
- Simple and predictable
- Large dense models are slow

MIXTURE-OF-EXPERTS

Only some experts per token

- Many experts, a few active per token
- Large total size, smaller active size
- Quality near a large model
- Lower cost and latency per token

THE TIEBREAKER

When the numbers show a mixture-of-experts model matching a larger dense model's quality at lower latency and cost, it wins.

Choosing size and architecture from the numbers

CANDIDATE	JUDGE PASS RATE	P95 LATENCY	RELATIVE COST
Dense, 8 billion	0.78	0.6 s	1x
Dense, 70 billion	0.88	1.9 s	6x
Mixture-of-experts	0.87	1.1 s	3x
Reasoning model, high effort	0.90	6.5 s	9x
Requirement	at least 0.85	under 1.5 s	lowest that passes

ON THE EXAM

Only the mixture-of-experts model passes both thresholds. The best score and the smallest model each fail one.

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Beyond pass or fail

- Quality usually shows **diminishing returns**: each extra point costs more.
- Compare **cost per quality point**, not just the top score.
- Reasoning effort is a dial: raise it only where the task needs it.
- Test with realistic load, since latency changes under concurrency.
- A benchmark from a model card is not your metric. Measure on your own set.

> ON THE EXAM
>
> The best choice usually sits at the knee of the curve, where more spending stops buying much quality.

The key metrics for each kind of application

SCENARIO	WATCH
Customer chat	Time to first token, p95, safety, groundedness, thumbs, escalations
Batch extraction	Field accuracy, failed rows, throughput, cost per thousand
Rag over documents	Retrieval relevance, groundedness, I-don't-know rate
Agent with tools	Task success, tool error rate, steps and cost per task
Code assistant	Suggestions accepted, tests passing, latency

ON THE EXAM

Match metrics to how the application fails. Chat fails slow or unsafe; batch fails wrong or dropped; agents fail unfinished.

Operational metrics, or quality metrics?

OPERATIONAL

Is it running well?

- Latency, errors, throughput, cost
- From usage and inference tables
- Cheap, on every request
- Blind to wrong answers

QUALITY

Is it answering well?

- Judges, task metrics, feedback
- From MLflow monitoring
- Sampled, costlier
- Blind to slowness and cost

THE TIEBREAKER

Monitor both. A fast, cheap application that answers wrongly is failing, and so is an accurate one nobody waits for.

Harbourline picks its metrics

01
Model
MoE chosen: passes quality and p95 targets.

02
Chat
TTFT, p95, safety, groundedness, escalations.

03
Claims batch
Field accuracy, failed rows, cost per 1,000.

04
Claims agent
Task success, tool errors, steps per task.

05
Alerts
Thresholds on p95 and groundedness.

ON THE EXAM

Different applications, different thermometers, one monitoring platform.

> Users of a chat assistant complain about slow replies. The dashboard shows an average latency of 1.2 seconds, which meets the target.
>
> Further analysis shows that 1 request in 20 takes more than 9 seconds.

Which metric should the team target and alert on?

A. The 95th percentile latency, which shows the slow tail

B. The average latency, since it already meets the target

C. Tokens per second, averaged across every request made

D. The total number of requests served per day by the endpoint

◆ PAUSE AND DECIDE

Which metric should the team target and alert on?

A p95 latency

One in twenty is exactly the 95th percentile tail.

B Average latency

Hides the tail that users are complaining about.

C Average tokens per second

Another average, and speed of generation, not waits.

D Requests per day

Volume, not how long anyone waits.

THE SIGNAL

Complaints while the average looks fine. One request in twenty is slow: that is precisely what p95 measures.

> Four candidates: dense 8B, pass rate 0.78, p95 0.6 s. Dense 70B, 0.88, 1.9 s. Mixture-of-experts, 0.87, 1.1 s. Reasoning model at high effort, 0.90, 6.5 s.
>
> The requirement is a pass rate of at least 0.85 and a p95 under 1.5 seconds, at the lowest cost.

Which model should the team choose?

A The dense 8B model, since it is the fastest and cheapest

B The dense 70B model, since it is the largest dense option

C The mixture-of-experts model, since it passes both limits

D The reasoning model, since it has the highest pass rate

◆ PAUSE AND DECIDE

Which model should the team choose?

A Dense 8B
0.78 fails the quality threshold.

B Dense 70B
1.9 s fails the latency threshold.

C Mixture-of-experts
0.87 and 1.1 s: the only candidate passing both.

D Reasoning model
6.5 s fails latency by a wide margin.

THE SIGNAL

Thresholds first. The best score and the smallest model each fail one, and only the mixture-of-experts model passes both.

> A nightly job uses ai_query to extract renewal dates, liability caps and governing law from thousands of new contracts into a table.
>
> The team is deciding what to monitor.

Which metrics matter most for this scenario?

A Time to first token and p95 latency on each request

B Field accuracy, failed-row count, and cost per thousand

C User thumbs-up rate and the average conversation length

D Safety judge pass rate on every extracted field value

◆ PAUSE AND DECIDE

Which metrics matter most for this scenario?

A TTFT and p95

Nobody is waiting per row in a nightly batch.

B Field accuracy, failed rows, cost

How a batch extraction job actually fails.

C Thumbs and conversation length

There are no users or conversations.

D Safety on fields

Dates and caps are not a safety risk.

THE SIGNAL

Nightly, batch, fields into a table. It fails by wrong fields, dropped rows and cost, so watch those.

> An agent books medical appointments by calling scheduling tools. Some patients report that bookings they asked for never appeared.
>
> The team wants metrics that would catch this automatically.

Which metrics should it monitor?

A Average response length, in tokens, across all replies

B BLEU score against a fixed set of reference replies

C Perplexity of the model on the booking conversations

D Task success rate, tool error rate and steps per task

◆ PAUSE AND DECIDE

Which metrics should it monitor?

A Response length

A polite reply can hide a failed booking.

B BLEU

Measures wording overlap, not whether a booking happened.

C Perplexity

Fluency, not task completion.

D Task success, tool errors, steps

Catches bookings that never completed.

THE SIGNAL

Bookings never appeared. An agent fails by not finishing its task, so measure task success and tool errors.

What to carry forward

- Five families: **quality, speed, cost, reliability, safety**.
- Users feel the tail: target **p95**, and time to first token for chat.
- **Mixture-of-experts** can match larger dense quality at lower latency and cost.
- Thresholds first, then cost. The best score and the smallest model often both fail.
- Match metrics to failure: chat, batch, rag, agent. Monitor operational and quality.

> ON THE EXAM
>
> Next: module 34 covers monitoring and cost control in production: inference tables, the gateway and alerts.